

Module 4

Statistics & Probability Foundations for Machine Learning

1. Understand Mathematical Foundations

- a) Variables and equations
- b) Functions
- c) Graphs
- d) Relationships between variables
- e) Mathematical thinking for ML

2. Understand Statistics Fundamentals

- a) Mean
- b) Median
- c) Mode
- d) Variance
- e) Standard deviation

3. Understand Probability Fundamentals

- a) Probability concepts
- b) Independent events
- c) Conditional probability
- d) Bayes' intuition
- e) Real-world probability examples

4. Understand Data Distributions

- a) Normal distribution
- b) Skewed distributions
- c) Outliers
- d) Variability in data
- e) Distribution interpretation

5. Understand Correlation & Relationships

- a) Positive correlation
- b) Negative correlation
- c) Correlation vs. causation
- d) Feature relationships
- e) Interpreting datasets

6. Understand Statistical Thinking for ML

- a) Sampling concepts
- b) Population vs. sample
- c) Bias awareness
- d) Variance awareness
- e) Data-driven conclusions

7. Thinking with AI

- a) Use AI for:
- b) Formula explanations
- c) Statistical interpretation
- d) Probability reasoning
- e) Validate calculations
- f) Verify generated conclusions

8. Apply Through Hands-on Tasks

- a) Analyze datasets statistically
- b) Calculate probability-based scenarios
- c) Visualize distributions
- d) Interpret statistical findings
- e) Identify patterns and relationships
- f) Explain insights using data